

Atharva Prabhu

Email — +1 716-275-9518 — LinkedIn — Portfolio — Open to Relocate

ML/AI Engineer with expertise in building end-to-end machine learning systems from research to deployment. Proven track record developing computer vision pipelines, NLP systems, and automated ML workflows with measurable impact. Strong foundation in deep learning, PyTorch, and MLOps with experience scaling models across production environments.

Education

MS Computer Science — University at Buffalo, SUNY — Aug 2024 – Dec 2025 — **GPA: 3.83/4.0**

Key Coursework: Deep Learning (A), Machine Learning (A), Pattern Recognition (A-), Operating Systems (A)

BE Computer Science — Savitribai Phule Pune University — Aug 2019 – July 2023

Professional Experience

AI/ML Intern — *Filmic Technologies* — Aug 2025 – Dec 2025

- Architected end-to-end automated film digitization quality analysis pipeline using Snakemake workflow orchestration, processing archival film footage across 13 specialized ML models for comprehensive quality assessment and metadata extraction
- Developed PyTorch-based computer vision models for sprocket hole pattern detection (1R vs 2R), aspect ratio identification, and optical soundtrack recognition, achieving automated technical analysis of digitized film characteristics
- Integrated Synchformer deep learning model for audio-visual synchronization offset detection, combining with Whisper ASR and Tesseract OCR for comprehensive transcription and audio quality analysis
- Engineered advanced image preprocessing pipeline implementing Gray World color correction, MSRCR enhancement, CLAHE, and edge enhancement to optimize degraded archival footage for ML inference
- Built full-stack deployment solution with React frontend and MySQL database backend, enabling automated database ingestion of detection results with real-time quality monitoring and metadata cataloging

Graduate Research Assistant — *A2IL Lab, University at Buffalo* — Ongoing

- Implemented real-time computer vision pipeline using YOLOv8 for object detection and NanoTrack for continuous tracking on Unitree Go2 quadruped robot, achieving robust ball detection under varying lighting conditions
- Integrated vision-to-motion control system converting detected ball coordinates into smooth robotic trajectories using PID control and gait optimization through Unitree Go2 SDK
- Contributed to ROS-based architecture enabling low-latency communication between vision processing and motion control modules for autonomous navigation

Bioinformatics Research Intern — *National Centre for Cell Science, Pune* — Aug 2023 – July 2024

- Developed GANGA pipeline for bacterial genome analysis using Snakemake, reducing computational time from 18 to 4 hours (78% improvement) and analyzed over 1,000 bacterial genomes
- Contributed to discovery of novel bacterial species (*Marinospirillum alkalitolerans*) resulting in publication in *Antonie van Leeuwenhoek* through comprehensive taxonomic classification and comparative genomics analysis
- Developed interactive dashboards using Anvi'o framework for automated classification of 10,000+ genomic datasets, delivering biological insights through data mining and statistical analysis for 7-member interdisciplinary team

Embedded Systems Development Intern — *Indio Networks Pvt Ltd, Pune* — July 2022 – Oct 2022

- Engineered automated deployment solutions using Python, reducing router firmware installation time by 60%

Technical Projects

Energy Demand Forecasting System — *Full-Stack MLOps Platform*

- Architected end-to-end MLOps platform for US electricity demand forecasting, orchestrating automated data pipelines, continuous model training, and real-time monitoring achieving MAE of 4,800 MWh on 24-hour forecasts with zero manual intervention
- Engineered automated ETL pipeline using GitHub Actions executing every 6 hours, integrating EIA API and Open-Meteo API to process 720 hourly records across 30-day rolling windows with 20+ engineered features and less than 1% error rate
- Implemented LightGBM regression model with automated hyperparameter optimization, achieving R^2 score of 0.54 and MAPE of 17-18% with less than 50ms inference latency and 95% confidence intervals for risk assessment
- Developed data drift detection system using Evidently AI with 30% threshold triggers, enabling automated model retraining and champion model selection through MLflow experiment tracking hosted on DagsHub
- Deployed REST API using FastAPI with Pydantic validation, Prometheus metrics export, and health check endpoints, serving real-time predictions with 99%+ uptime
- Designed CI/CD pipeline with 4 daily automated workflows handling data ingestion, drift monitoring, model retraining, and Git-based data versioning, achieving 100% automation and \$0 infrastructure cost

Computer Vision - Generative AI for Image Translation — *Advanced Deep Learning Project*

- Developed Pix2Pix cGAN system for sketch-to-photo translation, achieving final generator loss of 3.68 and SSIM score of 0.86 on facial image generation
- Engineered UNet encoder-decoder architecture with 8-layer downsampling/upsampling and strategic skip connections, enabling preservation of fine-grained sketch details in 256×256 output images
- Implemented PatchGAN discriminator framework with 30×30 patch-level classification and 70×70 receptive fields, achieving final discriminator loss of 0.0002 for robust adversarial training
- Built comprehensive data augmentation pipeline expanding CUHK dataset from 188 to 1,504 paired samples, implementing preprocessing with grayscale conversion and $[-1, 1]$ normalization

Technical Skills

Programming: Python, C/C++, R, SQL, BASH/Linux, Git

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, LangChain, Hugging Face Transformers

MLOps & Deployment: Docker, MLflow, FastAPI, Streamlit, Flask, DVC, Weights & Biases

Specialized Skills: Computer Vision (YOLO, ViT, GANs), NLP (BART, RAG, LLMs), Time Series Forecasting, Model Optimization

Big Data: Apache Spark, Hadoop, Distributed Computing, Workflow Orchestration (Snakemake)